9.13.4 The Access modifier const

const is the keyword used to declare a constant. You already learnt about constant in the previous chapter. const keyword modifies / restricts the accessibility of a variable. So, it is known as Access modifier.

For example,

int num = 100;

The above statement declares a variable num with an initial value 100. However, the value of num can be changed during the execution. If you modify the above definition as **const int num = 100;** the variable num becomes a constant and its value will remain 100 throughout the program, and it can never be changed during the execution.

```
#include <iostream>
using namespace std;
int main()
{
        const int num=100;
        cout << "\n Value of num is = " << num;
        num = num + 1; // Trying to increment the constant
        cout << "\n Value of num after increment " << num;
}
```

In the above code, an error message will be displayed as **"Cannot modify the const object"** in Turbo compiler and **"assignment of read only memory num"** in Dev C++.

9.13.5 References

A reference provides an alias for a previously defined variable. Declaration of a reference consists of base type and an & (ampersand) symbol; reference variable name is assigned the value of a previously declared variable.

Syntax:

<type> <& reference_variable> = <original_variable>;

Illustration 9.9: C++ program to declare reference variable

```
#include <iostream>
using namespace std;
int main()
{
        int num;
        int &temp = num; //declaration of a reference variable temp
        num = 100;
cout << "\n The value of num = " << num;
cout << "\n The value of temp = " << temp;
}
```

The output of the above program will be

The value of num = 100

The value of temp = 100

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